
Technical Audit of the Decoder–Gravity Notes

Generic one-flavor $\mathcal{N} = 2$ SYK first

A correction memo for *Fragile Fortuity: The Gravity of BPS Overlaps in Supersymmetric SYK*

Bottom line. There is a good paper inside the notes, but the current gravity derivation is not valid. The safe core is: the decoder is a compressed occupation projector; the independent-projector Jacobi/Wachter model is a useful auxiliary benchmark; its Weingarten expansion and the stated genus-one resolvent are mathematically correct; the GUE adjacent-gap result and the sub-edge population are reproducible. The main corrections concern the attempted self-energy reduction, the interpretation of the soft tail, the import of two-flavor counting into the one-flavor model, and the claim that BPS projection trivializes super-JT dynamics.

Scope. This audit treats the generic one-flavor, $q = 3$ model. The two-flavor monotone tower should be moved to a separate control section.

Checks performed. Independent exterior-algebra exact diagonalization, symbolic unitary Weingarten expansion through the sixth moment, and comparison with the LMRS zero-energy projected-operator calculation.

1 What was independently reproduced

The attached Mathematica source is readable. As attached, however, every executable line is wrapped in a comment, so the file does nothing if evaluated directly. Removing one outer comment layer recovers the intended program. The version supplied here uses 100 realizations, whereas Figure 13 of Ref. [1] used 500. There is also one reporting typo: the line labelled “largest $1 - \lambda$ ” takes a **Min** over realizations and should take a **Max**.

I ported the core calculation independently. For target size $N = 12$ and charge $p = 6$, every generic realization gave

$$\dim \mathcal{B} = 486, \quad n_0 = 80, \quad n_1 = 80, \quad n_{\text{int}} = 326.$$

Pooling 100 realizations gives 32,400 reduced adjacent-gap ratios and

$$\langle r \rangle = 0.5962 \pm 0.0025,$$

where the error is the standard error of the realization means. This is consistent with the published 0.599 ± 0.001 and with the large-matrix GUE mean 0.5996. The analytic Atas ratio surmise has mean 0.60266; these are two different benchmarks and should be labelled separately.

For target $N = 13$, $p = 6$, the auxiliary Wachter lower edge is $\lambda_- = 0.01293$. Five independent realizations gave 65–72 of 688 interior eigenvalues below this edge, i.e. 9.45%–10.47%, and a deep-tail fraction 4.22%–5.09% below $\lambda_-/5$. Thus the sub-edge population itself is real and easy to reproduce at low statistics.

The unitary Weingarten formulas in the notes were also checked symbolically. The planar moments and the D^{-2} coefficients agree through $k = 6$, including the proposed closed genus-one resolvent. This validates a statement about the *auxiliary Haar projection model*; it does not validate its identification with super-JT gravity.

2 Section 1: decoder and microscopic reduction

2.1 The decoder definition is sound

For one added mode, the basic operator

$$O_{\downarrow} = P_{\mathcal{B}}(1 - n_N)P_{\mathcal{B}}, \quad 0 \leq O_{\downarrow} \leq 1,$$

is correct. Its eigenvalues are squared singular values of the component map and squared cosines of principal angles between the BPS space and the unoccupied slot. The nonzero spectra of $D^{\dagger}D$ and $DD^{\dagger}$ coincide. These statements can remain.

Two qualifications are needed.

- The quantity called $\kappa = \int \lambda^{-1} d\mu - 1$ is infinite whenever the decoder has exact zero modes. The notes later quote finite values despite exact $\lambda = 0$ atoms. One must explicitly use a Moore–Penrose inverse, condition on the nonzero support, or return to the state-dependent dressing cost of Ref. [1].
- The trace mean is not generally b for each singled-out site and each realization. What is exact is the site sum

$$\frac{1}{N} \sum_{i=1}^N \frac{1}{d} \text{Tr}_{\mathcal{B}}[P_{\mathcal{B}}(1 - n_i)P_{\mathcal{B}}] = 1 - \frac{p}{N}.$$

For even N at half filling, particle–hole symmetry strengthens this to $\text{Tr } O_{\downarrow}/d = 1/2$ for each site and realization.

2.2 Equations (6)–(8) omit a harmonic component

The four block BPS equations are structurally correct, up to convention-dependent signs. The Hodge projector formula using the Green operator is also correct. The elimination that follows is not.

Let $u = D \wedge a$. From $Q^\dagger b = 0$ and $Qb = -u$ one obtains

$$Hb = -Q^\dagger u.$$

If $G = H^+$ is the Hodge Green operator, then the general solution is

$$b = b_0 - GQ^\dagger u, \quad b_0 \in \ker H.$$

The notes silently set $b_0 = 0$. Equivalently, they use $GH = 1$, whereas the exact identity is

$$GH = HG = 1 - P_B.$$

Consequently their norm calculation gives only the non-harmonic part,

$$\|b - b_0\|^2 = \langle a | C'^\dagger G C' | a \rangle,$$

not the full $\|b\|^2$. The fourth BPS equation is also still to be imposed; it both restricts a and fixes harmonic data.

Required deletion. The asserted exact formula

$$O_\downarrow = (1 + C'^\dagger G C')^{-1} \quad \text{on } \ker Q$$

is false for the physical decoder. It cannot produce exact zero eigenvalues, and it misses precisely the old-BPS harmonic component that dominates the observed lower tail. The phrases “exact self-energy form” and all deductions that rely on Eq. (8) should be removed unless a new constrained Schur-complement derivation is supplied.

A correct replacement is a constrained generalized eigenvalue problem. Write $R = D \wedge$. On pairs (a, b_0) satisfying

$$P_{\mathcal{B}_N^{p+2}} R a = 0, \quad Q^\dagger a + R^\dagger b_0 - R^\dagger Q^\dagger G R a = 0,$$

the decoder numerator is $\|a\|^2$, while the physical norm is

$$\langle a | (1 + R^\dagger G R) | a \rangle + \|b_0\|^2.$$

Equivalently, compress $A = \text{diag}(1, 0)$ and $M = \text{diag}(1 + R^\dagger G R, 1)$ to this constraint space and solve $Av = \lambda Mv$. This formulation keeps the harmonic channel and permits both endpoint eigenvalues.

There is also a degree bookkeeping error in the prose: in $b = -GQ^\dagger u$, G acts at the degree of b ; after commuting G through $Q^\dagger$, the equivalent quadratic form uses G at the degree of u .

3 Sections 1.3–1.4: Wachter law and endpoint modes

3.1 Keep Wachter as an auxiliary null model

For two independent projections of ranks $d = aD$ and $m = bD$, the normalized spectrum of PQP on the d -dimensional P -space is indeed the free Jacobi/Wachter measure, with

$$\lambda_\pm = \left(\sqrt{a(1-b)} \pm \sqrt{b(1-a)} \right)^2$$

and the stated endpoint atoms. The first three planar moments quoted in the notes are correct. What is not established is the premise that the physical BPS projector is independent Haar relative to the selected occupation slot. The new couplings are independent of the old couplings, but the *enlarged* BPS subspace is defined using both. Supersymmetry confines its two components to $\ker Q$ and $\ker Q^\dagger$, creating exact intersections and correlations. The numerical result is more precise:

The conditional interior density is often close to Wachter, but the full physical measure is not an independent-projection measure. At $N = 12, p = 6$, the physical endpoint multiplicities are $80 + 80$, whereas independent projections with the same $(D, d, m) = (924, 486, 462)$ have only $24 + 24$ forced endpoint modes.

Any histogram of only interior eigenvalues must compare with the *conditional* continuous Wachter density, divided by its continuous mass. The notes should state this normalization explicitly.

The interval $[\lambda_-, \lambda_+]$ is the planar/free-probability support, not a strict finite- D exclusion interval. A finite Jacobi matrix can fluctuate past this edge. The meaningful comparison is quantitative: ordinary Haar edge fluctuations are tiny, whereas the observed physical population below λ_- is about ten percent at the sizes tested. Thus “Haar gives exactly zero sub-edge eigenvalues” should be replaced by a finite-sample statement.

The claim below Figure 2 that agreement “tightens with N ” is also not supported by the displayed KS values 0.111, 0.060, 0.112, 0.046, 0.101. They are non-monotonic; the safe statement is simply that the conditional bulk is reasonably close over the displayed sizes.

3.2 The protected-index paragraph mixes two different models

The characterization of exact endpoints is correct:

$$\lambda = 1 : \quad a \in \mathcal{B}_N^p \cap \ker(D\wedge), \quad \lambda = 0 : \quad b \in \mathcal{B}_N^{p-1} \cap \ker \iota_{\bar{D}}.$$

Their multiplicities in the generic one-flavor model are *not* the two-flavor monotone-tower number $3^{(N-1)/2}$. Equation (14) therefore mixes models and also adds a normalized density to an absolute state count. Delete it and report the one-flavor kernel dimensions separately. The two-flavor tower belongs in a later control section.

At even target N and $p = N/2$, there is an additional exact statement absent from the notes. The antiunitary Hodge/particle-hole map sends $Q \leftrightarrow Q^\dagger$, preserves the BPS space, and sends $n_i \leftrightarrow 1 - n_i$. Hence

$$\mathcal{C}O_\downarrow \mathcal{C}^{-1} = 1_{\mathcal{B}} - O_\downarrow.$$

The spectrum is paired as $\lambda \leftrightarrow 1 - \lambda$ in each realization, the endpoint multiplicities are equal, and all odd centered moments vanish. The numerical pairing residual at $N = 8, 10, 12$ is below 3×10^{-15} .

4 Sections 2–3: Haar genus expansion

The Weingarten identity is correct for the auxiliary Haar model. At fixed a, b it has a unitary topological expansion in even powers of D^{-1} . The quoted coefficients $m_k^{(1)}$ for $k = 2, 3, 4$ are correct, and the proposed

$$W_1(x) = -\frac{b(1-a)(1-b)x(x-1)}{[(x-\lambda_-)(x-\lambda_+)]^{5/2}}$$

reproduces the exact Weingarten coefficients through $k = 6$.

There are nevertheless three corrections.

1. At $a = b = 1/2$, the notes state $m_2 = 3/8 - (1/4)D^{-2} + \dots$. The correct expansion is

$$m_2 = \frac{3D^2/8 - 1/2}{D^2 - 1} = \frac{3}{8} - \frac{1}{8D^2} + O(D^{-4}).$$

This also agrees with the general coefficient $-b(1-a)(1-b) = -1/8$.

2. The phrase “the whole handle tower resums into W_1 ” is misleading. W_1 resums all one-boundary moments at *genus one*; it does not resum all genera.
3. The $b \rightarrow 0$ relation to Marchenko–Pastur requires an eigenvalue rescaling. Without rescaling, the mean eigenvalue tends to zero and the measure collapses at the origin.

The “four surfaces” for m_2 and the $12 + 21 + 3$ count for m_3 are valid as leading-genus classifications of permutation terms. They should be called Haar fatgraphs unless and until an actual super-JT boundary condition is derived.

The normalization of W_1 need not be “pinned at six points.” Its overall factor follows analytically by matching the exact $m_2^{(1)}$ coefficient. Replace the fit language with that one-line derivation.

5 Section 4: the gravity derivation does not work

This is the most important conceptual correction.

5.1 BPS projection does not trivialize the super-Schwarzian

LMRS define projected operators $\hat{O} = POP$ with $P = \lim_{\beta \rightarrow \infty} e^{-\beta H}$ and compute their traces from long-time correlators. Even though the external states have zero energy, the answer is a nontrivial super-Schwarzian/super-Liouville propagator calculation and depends on the operator dimension, charge, and normalization [2]. Projecting external boundaries to the BPS sector therefore does *not* leave only e^{S_0} .

Delete the statements that a kinematic BPS observable “cannot depend on Schwarzian dynamics,” that every boundary simply contributes the BPS delta-function weight, and that Eq. (29) is the complete amplitude. These statements contradict the LMRS calculation that this project is supposed to use.

5.2 The entropy identification is inconsistent with super-JT

In $\mathcal{N} = 2$ super-JT, e^{S_0} controls the zero-energy black-hole degeneracy (with charge-dependent factors). In the SYK comparison of LMRS, the BPS count is $N_{\text{BPS}} = e^{S_0} \hat{L}$. The notes instead set

$$e^{S_0} = D = \binom{N}{p},$$

the full ambient Fock-sector dimension, while treating the actual BPS count $d \sim 3^{N/2}$ as a brane-flavor number. These quantities have different entropy densities, $\log 2$ and $\frac{1}{2} \log 3$. No derivation is given for replacing the gravitational extremal entropy by the ambient Hilbert-space entropy. Thus the proposed D^{-2} Haar genus parameter cannot presently be identified with the super-JT handle weight.

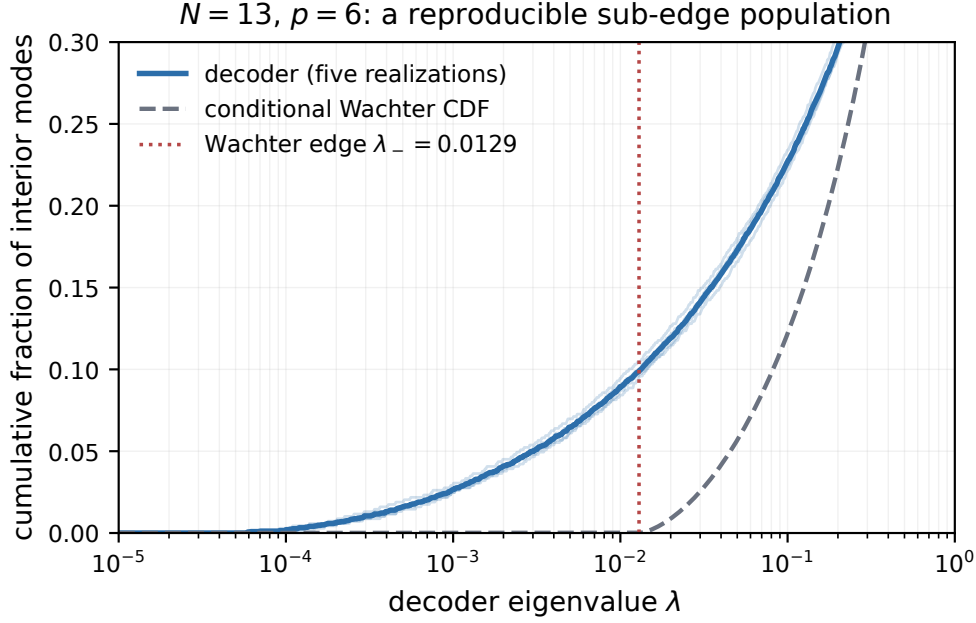


Figure 1: Independent low-statistics reproduction of the sub-edge population. The five thin curves are individual realizations; the solid curve is their mean.

5.3 The second EOW-brane species is postulated, not derived

The factorization $O_{\downarrow} = D^{\dagger}D$ makes it a Gram operator in elementary linear algebra. It does not imply that the rows of D are physical EOW-brane states, nor that the slot rank m is a brane-flavor multiplicity. A valid EOW construction must derive the boundary condition representing $1 - n_i$, its normalization, and its super-JT amplitudes. Until then, the PSSY/Wachter discussion should be labelled an auxiliary model or conjecture.

For the same reason, “the action reproduces the handles to all orders” is unsupported. What is exact is only the equivalence between the Haar integral, its Weingarten expansion, and its fatgraph/topological-recursion organization.

6 Sections 5–6: soft tail

6.1 The tail exists, but the moment statement is impossible

On the compact interval $[0, 1]$, the full sequence of moments determines the probability measure. Therefore a measure cannot have exactly the same positive moments as Wachter while differing by a macroscopic $O(10\%)$ tail. Replace “reproduces every positive moment” with the empirical statement that low moments and the in-band bulk can be close to the Haar benchmark.

The observations at $N = 12$ – 15 establish a finite-size sub-edge sector, not its large- N fate. Four sizes do not prove monotonic asymptotic growth, saturation at a nonzero fraction, or an $O(1)$ takeover. The fitted small- λ exponents are likewise too size-dependent to support a universal power law.

6.2 Equation (32) is exact, but E_b is not an energy eigenvalue

For a decoder eigenvector $\Psi = (a, b)$, the norm balance gives

$$\lambda = \frac{E_b}{E_b + \sigma_a^2}, \quad E_b = \frac{\langle b | H_N | b \rangle}{\|b\|^2}, \quad \sigma_a^2 = \frac{\|D \wedge a\|^2}{\|a\|^2}.$$

This identity was verified to machine precision. However, E_b is a Rayleigh quotient of a component of a decoder eigenvector. It is not generally an eigenvalue of H_N . The claimed one-to-one map to the near-BPS energy spectrum and the density push-forward

$$\rho_{\text{tail}}(\lambda) = c^{-1} \rho_H(\lambda/c)$$

do not follow.

A direct diagnostic shows the problem. Decompose the occupied component into old-theory Hodge pieces,

$$b = b_0 + b_\perp, \quad b_0 \in \ker H_N, \quad \epsilon = \frac{\|b_\perp\|^2}{\|b\|^2}.$$

Then, exactly,

$$E_b = \epsilon \bar{E}_\perp, \quad \lambda = \frac{\epsilon \bar{E}_\perp}{\sigma_a^2 + \epsilon \bar{E}_\perp}, \quad \bar{E}_\perp = \frac{\langle b_\perp | H_N | b_\perp \rangle}{\|b_\perp\|^2}.$$

In the five $N = 13, p = 6$ samples, the median harmonic fraction of a tail vector is 99.57%–99.80%, while the median $\bar{E}_\perp$ is an ordinary 16–29 in the chosen normalization. The small Rayleigh quotient is caused by a tiny *weight* outside the old BPS subspace, not by localization on an individual near-zero energy eigenmode.

Correct interpretation. The soft lower tail is a family of target BPS states lying very close to the occupied copy of the old BPS subspace. It is a near-intersection/principal-angle phenomenon. The upper tail has the particle–hole mirror interpretation. This is stronger and cleaner than the unsupported energy-density push-forward.

The fitted constant c is consequently not a theorem and not presently predicted from the couplings. It packages the state-dependent quantities $\bar{E}_\perp/\sigma_a^2$.

7 Section 7: large- N corrections

The central-sector BPS dimensions in Eq. (36) are interchanged by a factor of $3/2$. For generic $q = 3$ one-flavor SYK,

$$d_N = \begin{cases} 2 \cdot 3^{N/2-1}, & N \text{ even}, \quad p = N/2, \\ 3^{(N-1)/2}, & N \text{ odd}, \quad p = (N-1)/2 \text{ or } (N+1)/2. \end{cases}$$

Thus $d_{12} = 486$, not 729. The entropy density $\frac{1}{2} \log 3$ is still correct, but the parity-dependent prefactors and the displayed asymptotic prefactor for $a = d/D$ must be fixed.

For the *auxiliary* Haar model, $a \rightarrow 0$ and $b \rightarrow 1/2$ do imply

$$\lambda_\pm = \frac{1}{2} \pm \sqrt{a} + O(a^{3/2}),$$

so its continuous support collapses to a delta function at $1/2$. This does not mean that the bulk carries vanishing spectral weight: in the Haar model the delta spike carries unit weight once the endpoint atoms vanish. Delete the claims that “the bulk disappears” and that the Haar/wormhole description captures a vanishing fraction. The physical fraction away from $1/2$ is an open problem and cannot be inferred from support width alone.

8 Section 8 and appendices

- Distinguish the analytic GUE ratio surmise mean 0.60266 from the large-matrix numerical GUE mean 0.5996(1). Both are standard; the notes should label which benchmark is being used.
- Separate the one-flavor chaotic decoder from the two-flavor deterministic monotone tower. Their contrast is useful, but they are not pieces of one spectral decomposition unless a common model is explicitly defined.
- The provisional statement that the sub-edge levels have GUE correlations is plausible, but should retain its stated finite-statistics caveat.
- Appendix A may keep the formula for W_1 , after replacing numerical normalization fitting by analytic matching to $m_2^{(1)}$.
- Appendix B.1 is exact only for the auxiliary Haar model. It is not the exact disorder average of the physical decoder once the notes themselves exhibit non-Haar endpoint and tail structure.
- Appendices B.2–B.4 are a proposed gravitational interpretation, not a derivation. In particular, Eq. (48) omits the nontrivial matter/super-Schwarzian amplitudes computed by LMRS.

9 Recommended rewrite order

1. **Exact one-flavor SYK geometry:** decoder definition, block BPS equations, endpoint kernels, particle-hole pairing, and exact norm identities. Remove the false self-energy formula.
2. **Numerics:** GUE adjacent-gap statistics, conditional Wachter-like bulk, exact endpoint excess, and the reproducible soft tail. Replace the energy-spectrum map by the Hodge-distance interpretation.
3. **Auxiliary random-projection benchmark:** Wachter law, exact Weingarten moments, and the verified W_1 . State clearly that this is a null model, not yet gravity.
4. **LMRS calculation:** identify the centered decoder with a projected neutral bilinear and compute at least the second moment from super-JT. This calculation is carried out in the companion research note.
5. **EOW stage:** present only after deriving a boundary condition for the slot operator and resolving the $e^{S_0} = d$ versus D entropy mismatch. Until then, call the two-brane Wachter construction conjectural.
6. **Two-flavor control:** put the monotone tower and its protected counting in a separate final section.

The most publishable message is not that the entire decoder is already an all-genus topological gravity observable. It is that the same canonical operator simultaneously has: (i) exact cohomological endpoint structure, (ii) an approximately Jacobi one-point bulk, (iii) GUE local correlations, and (iv) moments that can be attacked directly by the LMRS super-JT machinery. That statement is both interesting and supportable.

References

- [1] J. Chryssanthacopoulos and D. Vegh, *Fortuity and fragility in supersymmetric SYK*, arXiv:2608.12160 (2026).
- [2] H. W. Lin, J. Maldacena, L. Rozenberg, and J. Shan, *Looking at supersymmetric black holes for a very long time*, SciPost Phys. 14, 128 (2023), arXiv:2207.00408.
- [3] Y. Y. Atas, E. Bogomolny, O. Giraud, and G. Roux, *Distribution of the ratio of consecutive level spacings in random matrix ensembles*, Phys. Rev. Lett. 110, 084101 (2013).